

Installation guide and instructions of INSTRON fast data analysis tool for the COWI clay project

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Introduction:

Extracting the data from the INSTRON machines of DTU and using it to produce plots can be cumbersome, as the data are printed in separate csv (or txt) files for each testrun, and each test series are collected in separate folders.

This program concatenates the csv files of these folders into single csv files, and produces plots or extracts relevant values from the test runs quickly.

The pre-requisites for the program being able to run are still somewhat cumbersome, as the naming convention and the file structure must be in a certain manner, but it reduces the workload of the actual data analysis heavily.

Requirements:

The program has been written in python 3.11, however as there are no functions used that have been deprecated thus far, it should run just as well on newer versions of python, assuming the modules of [requirements.txt](#) are compatible with the newest version of python as well.

Python can be installed either via the windows store or by going to python.org/downloads and choosing the desired version. Make sure that pip is also installed when installing python.

Installation:

To install it, download it from the github webpage. Extract the files to a folder wherever you want it, and right click in the folder and choose “open in terminal”. The commandline should display the path to the location of the folder. Run the command “python -m pip install -r requirements.txt”. If the error “The term ‘python’ is not recognized as the name of a...” is thrown, try running the command as “python3 -m pip install -r requirements.txt” instead, as the pathname of your python installation can vary depending on the installed python version.

Windows:

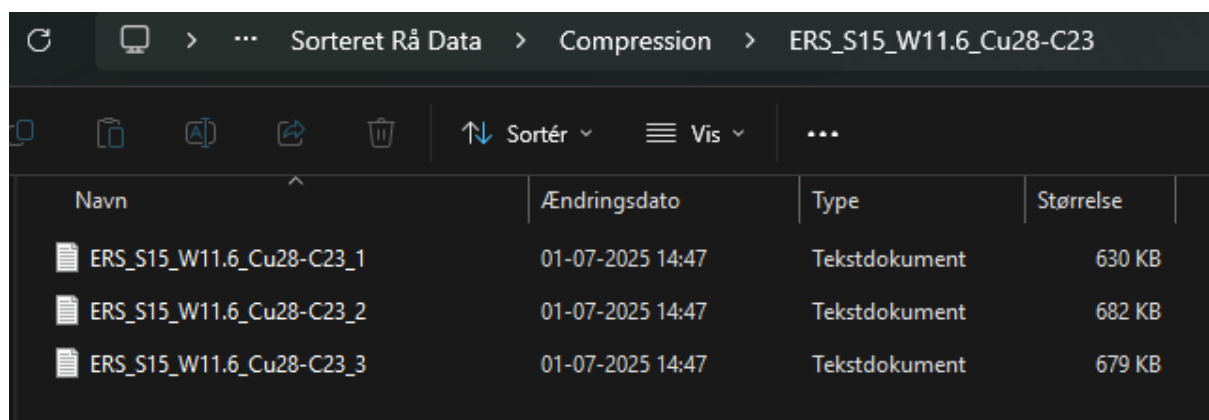
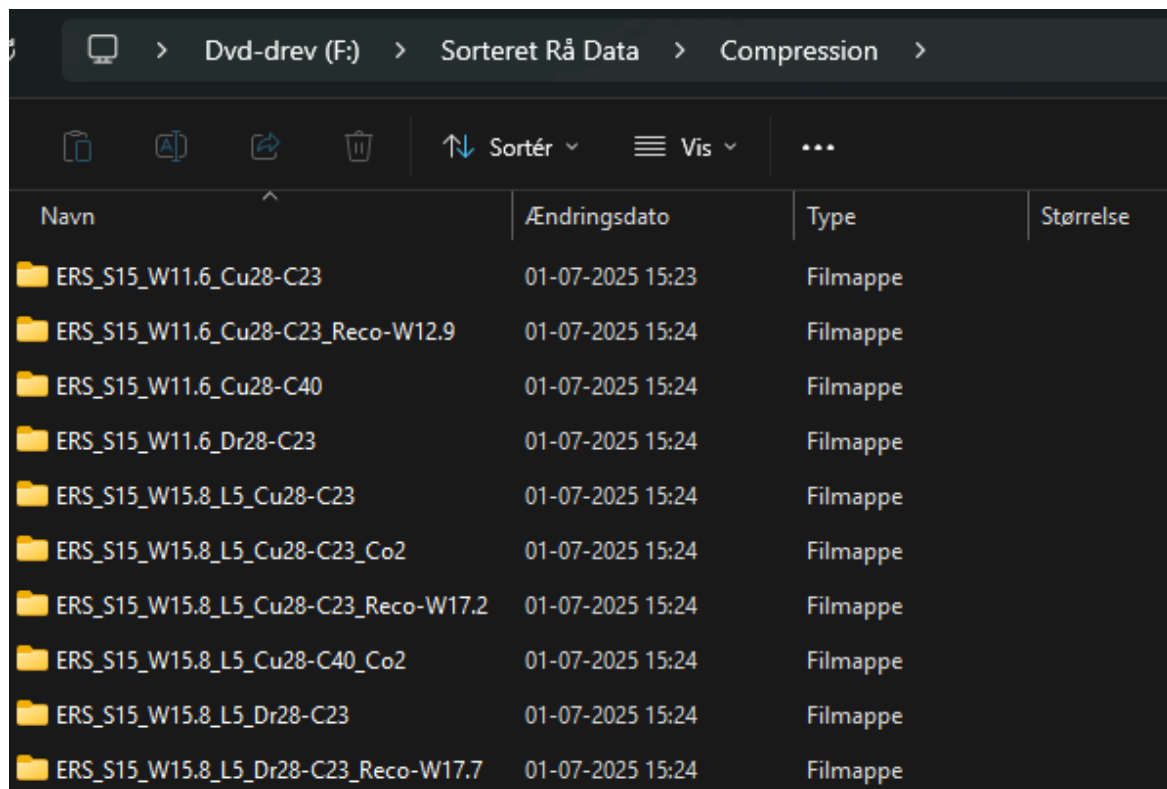
If pip successfully completes installation of all the packages, the program can be run by using “run.bat”

MacOS/Linux:

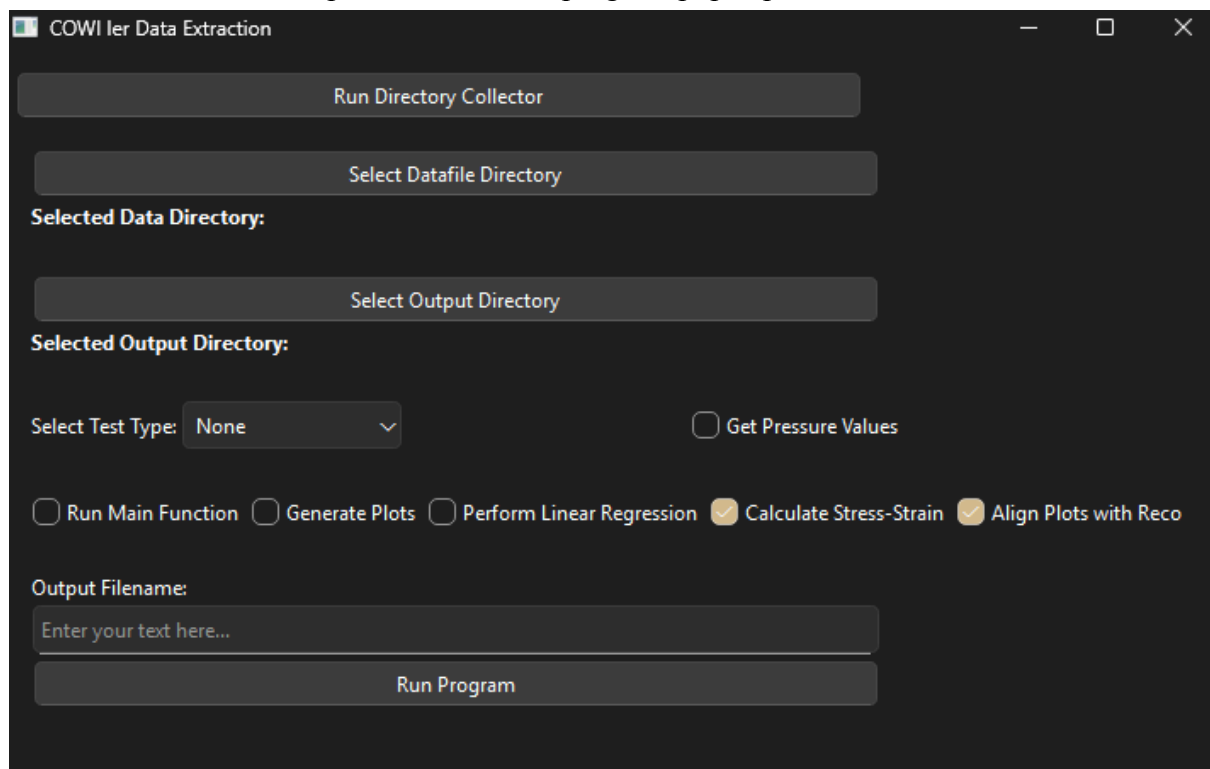
If pip successfully completes installation of all the packages, the program can be run by typing “python [app.py](#)” or “python3 [app.py](#)” again depending on the python pathname.

Running the program:

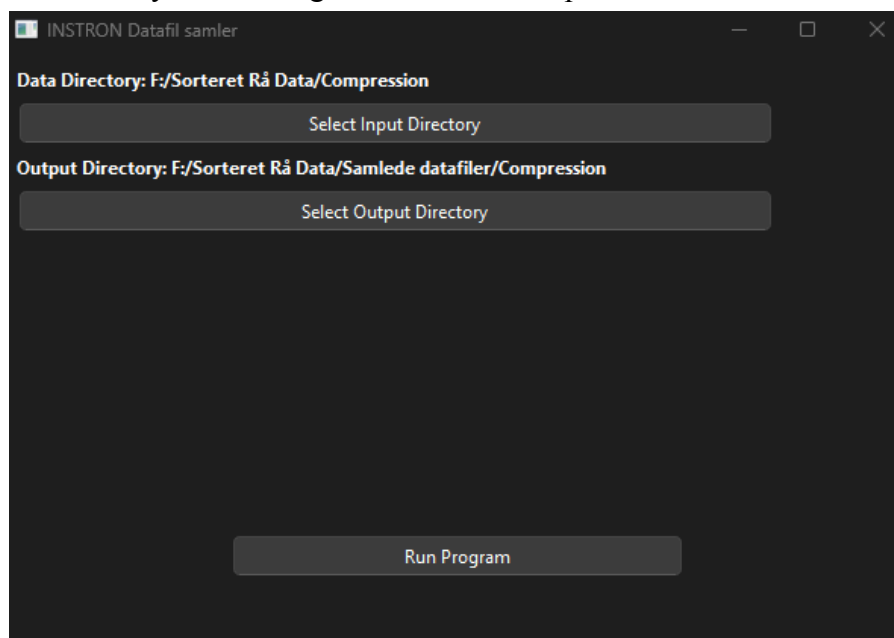
Let’s walk through how to use the program to get data for compression tests. I have a folder of folders, that each contain run data from compression testing, and I have a folder where i wish for the concatenated csv files to be placed.



When i run the batch script, the GUI of the program pops up:

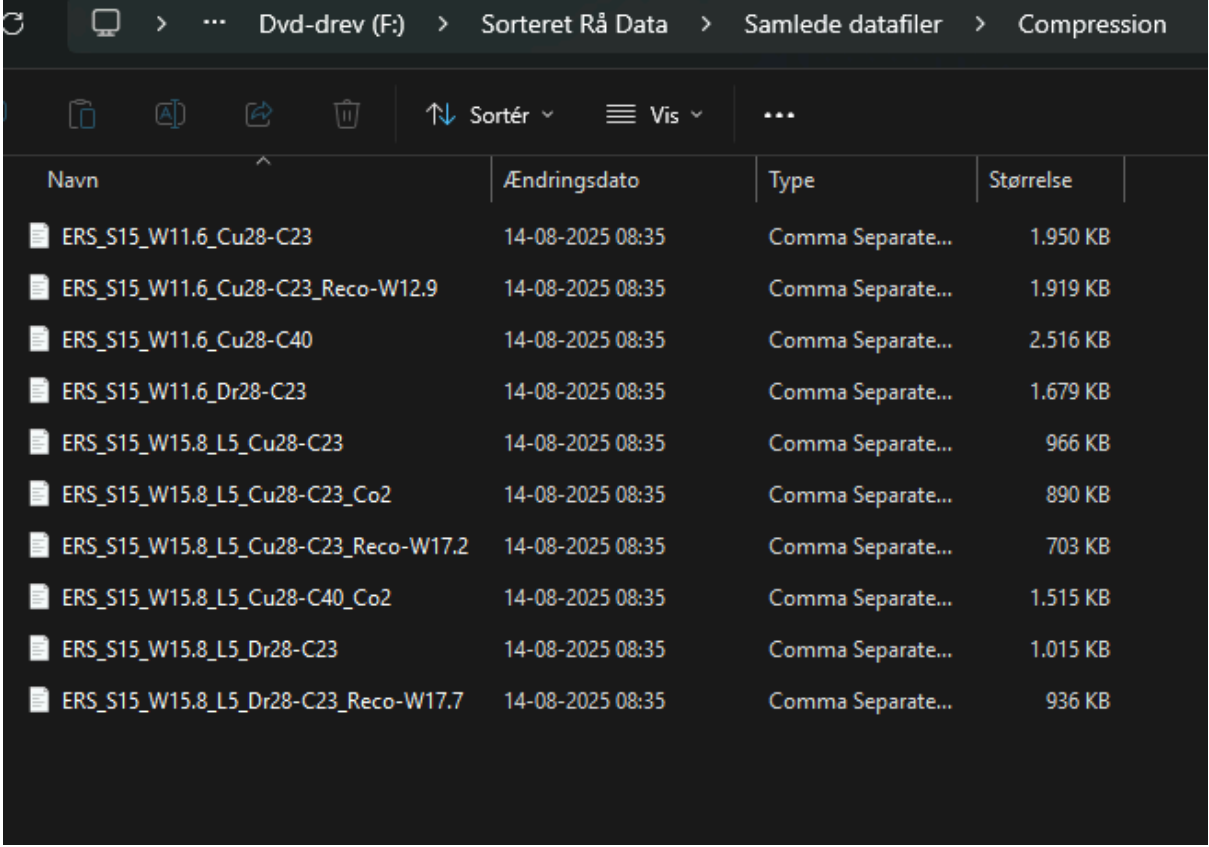


When i wish to collect Data files, I choose to run the directory collector. Be aware that any old files with identical names in the target folder will be overwritten. This is so that when you get new data you dont have to manually delete files from your target folder, you can just run the directory collector again and it will be updated.



After selecting the directories I press Run Program and my output directory gets filled with the CSV files which the main program can use for data analysis. It is possible to skip the step of producing these csv files and produce the plots directly from the raw data, however it would require refactoring a large portion of the code, so that is not something I have put energy into yet.

After running the program my folder is filled with csv files that retain the names of the test series:



Navn	Ændringsdato	Type	Størrelse
ERS_S15_W11.6_Cu28-C23	14-08-2025 08:35	Comma Separate...	1.950 KB
ERS_S15_W11.6_Cu28-C23_Reco-W12.9	14-08-2025 08:35	Comma Separate...	1.919 KB
ERS_S15_W11.6_Cu28-C40	14-08-2025 08:35	Comma Separate...	2.516 KB
ERS_S15_W11.6_Dr28-C23	14-08-2025 08:35	Comma Separate...	1.679 KB
ERS_S15_W15.8_L5_Cu28-C23	14-08-2025 08:35	Comma Separate...	966 KB
ERS_S15_W15.8_L5_Cu28-C23_Co2	14-08-2025 08:35	Comma Separate...	890 KB
ERS_S15_W15.8_L5_Cu28-C23_Reco-W17.2	14-08-2025 08:35	Comma Separate...	703 KB
ERS_S15_W15.8_L5_Cu28-C40_Co2	14-08-2025 08:35	Comma Separate...	1.515 KB
ERS_S15_W15.8_L5_Dr28-C23	14-08-2025 08:35	Comma Separate...	1.015 KB
ERS_S15_W15.8_L5_Dr28-C23_Reco-W17.7	14-08-2025 08:35	Comma Separate...	936 KB

I can now close the window for data directory collection, or use it again if i want to also collect the raw data from splitting, wet compression or production.

In the main GUI window it should now state “Process completed successfully!” at the bottom of the window. This means it is ready for Data analysis.

Here I have chosen the datafile directory to be the one we just filled with files. I have chosen the test type as Compression. I have chosen for it to not output the pressure values, meaning the datafile i get out will contain only the maximum of the forces, not the corresponding pascal pressure.

I have ticked “Run Main Function” which generates to csv files, one containing characteristics of the test series and the corresponding mean of the max forces with standard deviation for the series, and one containing the max force values for each of the tests in the series.

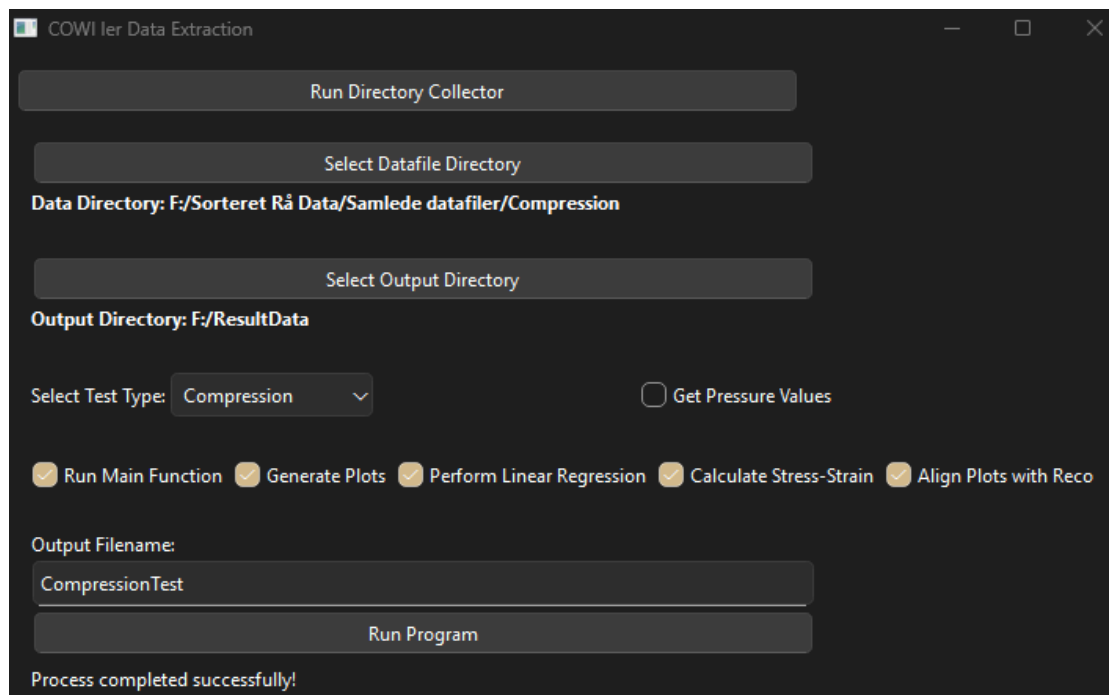
I have ticked Generate plots which produces a png file. If Perform Linear Regression is also ticked, there will also be a csv file generated containing values for the linear regression.

Currently, the linear regression is always performed in the 10-40% interval of the max force and the 40-80% interval of the max force. This is because many of the samples had a characteristic bend at almost half the maximum force of the sample, so we assume that it always happens at 40%. The code can be modified easily to allow for other percentage intervals to be used if that’s required. If Linear Regression is not ticked the raw data will be plotted and all the plots will be scaled after the highest value max force, meaning if the highest measured force of any test is 10 kN, all the plots will have ylims of 0-11 kN.

I have ticked calculate Stress-Strain which causes the plots to be stress-strain plots rather than force-displacement plots

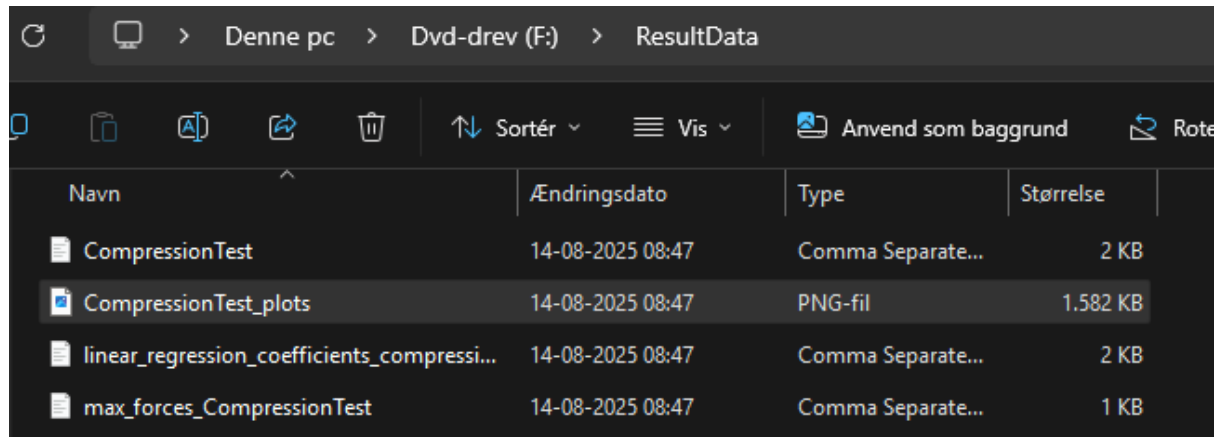
I have ticked Align plots with Reco which pairs a test with its recompressed version.

Currently indexing of plots is defined by a dictionary in [DataExtraction.py](#). This dictionary must be updated when naming convention changes. It is possible to choose between the two naming conventions only by modifying inside [DataExtraction.py](#), as I felt the GUI was already cluttered enough without all the extra options.



There are some combinations of settings which do not make sense. For example if perform linear regression is ticked but generate plots is not ticked, the program will not run. This is reported in the error handler beneath the Run Program button (which currently states Process completed successfully).

When running the program with these settings i get this:



All the CSV files can be opened in Excel, and be turned into regular cells by using the data function of excel. To do this, go to the “Data” section. Highlight the first column (column A). It looks like they are in more columns but they are all in the first column. Click the “Text to columns”. Choose the “Delimited” option and press next - Check the Delimiter to be “Comma” and uncheck “Tab” and click next - click finish